Trader Performance vs. Bitcoin Market Sentiment

Analysis of Hyperliquid historical trades against the Fear & Greed Index

1. Data & Methodology

Dataset	Rows	Range	Notes
fear_greed_index.csv	2,644 daily records	2018-02-01 → 2025-05-02	classification bucketed into 5 levels
historical_data.csv	211,224 trade fills	2023-05-01 → 2025-05-01	32 accounts, 246 coins; incl. Closed PnL, Size USD, Direction, Side, Fee

Join: trades were matched to the sentiment classification for their calendar date (IST). 211,218 of 211,224 trades (100%) matched a sentiment day.

Key metric definitions:

- *Closed PnL* — realized profit/loss per fill; only fills with non-zero Closed PnL (104,408 of them) were used for win-rate/avg-PnL statistics (zero-PnL rows are position-opening fills, not closes).
- *Win rate* — % of non-zero-PnL closes that were profitable.
- *Long/short bias* — % of all trade actions (not just closes) tagged as long-side vs. short-side via the Direction field.

All code, intermediate CSVs, and charts accompany this report in the delivered folder (`analysis.py` + 6 PNGs + supporting CSVs).

2. Headline Numbers by Sentiment Regime

Sentiment	Trades	Total PnL	Avg PnL/trade	Win Rate	Total Volume
Extreme Fear	10,406	$739,110	$71.03	76.2%	$56.9M
Fear	29,808	$3,357,155	$112.63	87.3%	$239.7M
Neutral	18,159	$1,292,921	$71.20	82.4%	$101.0M
Greed	25,176	$2,150,129	$85.40	76.9%	$136.9M
Extreme Greed	20,853	$2,715,171	$130.21	89.2%	$57.9M

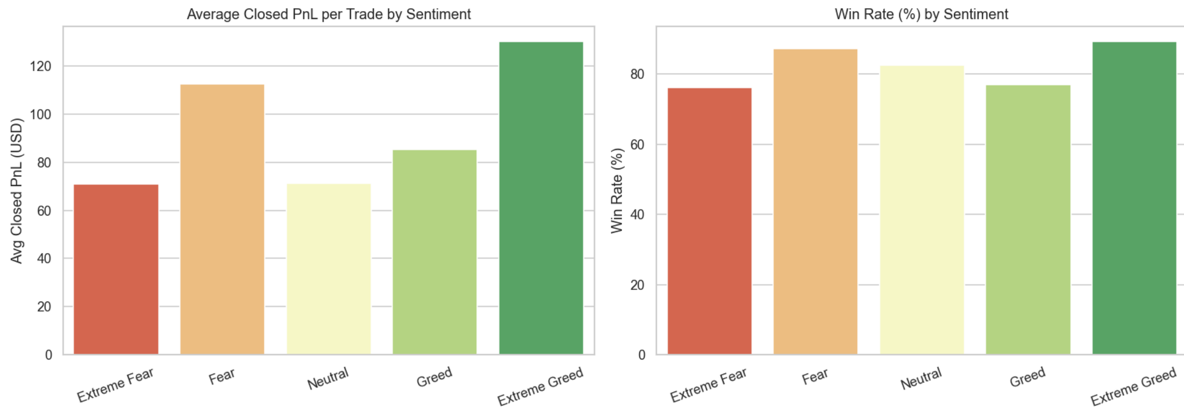

Figure 1. Average Closed PnL per trade and win rate (%) across the five sentiment regimes.

3. Key Findings

3.1 Traders are contrarian, not momentum-driven

Position direction flips almost perfectly with sentiment:

Sentiment	% Long actions	% Short actions	Avg size (USD)
Extreme Fear	65.7%	34.3%	$5,350
Fear	62.0%	38.0%	$7,816
Neutral	61.4%	38.7%	$4,783
Greed	42.3%	57.7%	$5,737
Extreme Greed	46.6%	53.4%	$3,112

This trader cohort is net **long-biased during Fear/Extreme Fear** and flips to **net short-biased during Greed** — i.e., buying the dip when the crowd is fearful and fading rallies when the crowd is greedy. This is a classic contrarian / “smart money” pattern, not a herd-following one.

Figure 2. Long vs. short action share by sentiment regime.

3.2 The two extremes are the most profitable; “Neutral” is the weakest

Both **Fear** and **Extreme Greed** produce the best win rates (87–89%) and highest average PnL per trade ($113 / $130). **Neutral** sentiment is the worst per-trade performer (71.2% avg PnL, tied with Extreme Fear) despite a decent win rate — consistent with low-conviction, choppy/range-bound markets being harder to extract edge from than emotionally extreme ones.

Extreme Fear, despite high activity, has the **lowest win rate (76.2%) and lowest avg PnL ($71)** of all buckets — traders are most active here but least accurate, suggesting fear days attract more speculative/panic entries.

3.3 Activity concentrates in Extreme Fear, but efficiency doesn't

On a per-day basis, Extreme Fear days see by far the most trades (1,529/day) and volume ($8.2M/day) — more than double any other regime — yet deliver the lowest daily win rate (65%). Traders over-trade during panic without a proportional payoff.

Sentiment	Avg daily PnL	Avg daily volume	Avg daily win rate	Avg trades/day
Extreme Fear	$52,794	$8.18M	65%	1,529
Fear	$36,892	$5.31M	88%	680
Neutral	$19,297	$2.69M	79%	562
Greed	$11,141	$1.50M	81%	261
Extreme Greed	$23,817	$1.09M	89%	351

3.4 Correlation with the numeric sentiment score (0–100)

Weak-to-moderate correlations across the board — sentiment level alone is not a strong linear predictor of daily outcomes:

Metric	Corr. with sentiment value
Total daily PnL	-0.08
Total daily volume	-0.26
Win rate	+0.06
Trade count	-0.25
Unique active traders	-0.28

The negative correlations with volume/trade-count/active-traders confirm §3.3: **lower sentiment scores (more fear) associate with more trading activity**, not more profit. Win rate and PnL show near-zero linear correlation with the raw score — the relationship is regime-based (U-shaped across the 5 buckets), not linear.

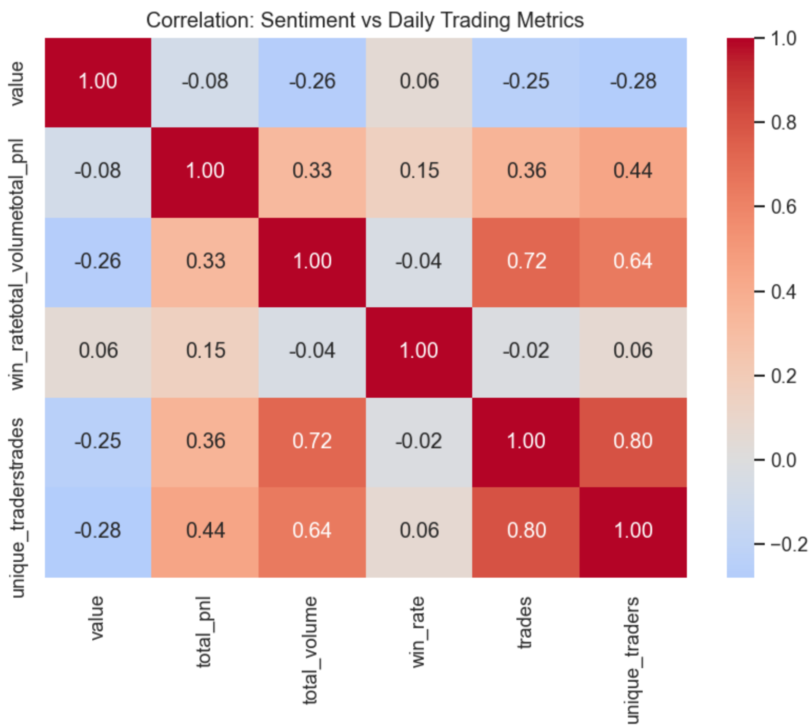

Figure 3. Correlation matrix between sentiment value and daily trading metrics.

3.5 Coin concentration is stable, but shifts slightly with sentiment

BTC dominates volume in every regime, but its dominance is strongest in Fear (BTC ≈ 55% of volume) and weakest in Extreme Fear / Extreme Greed, where HYPE and SOL take a larger share — consistent with capital rotating into higher-beta alt positioning at emotional extremes rather than sitting purely in BTC.

3.6 Position sizing shrinks as greed rises

Average position size peaks in Fear ($7,816) and steadily declines into Extreme Greed ($3,112) — traders here are sizing up in fear and de-risking (smaller size, more short activity) into greed, reinforcing the contrarian risk-management read from §3.1.

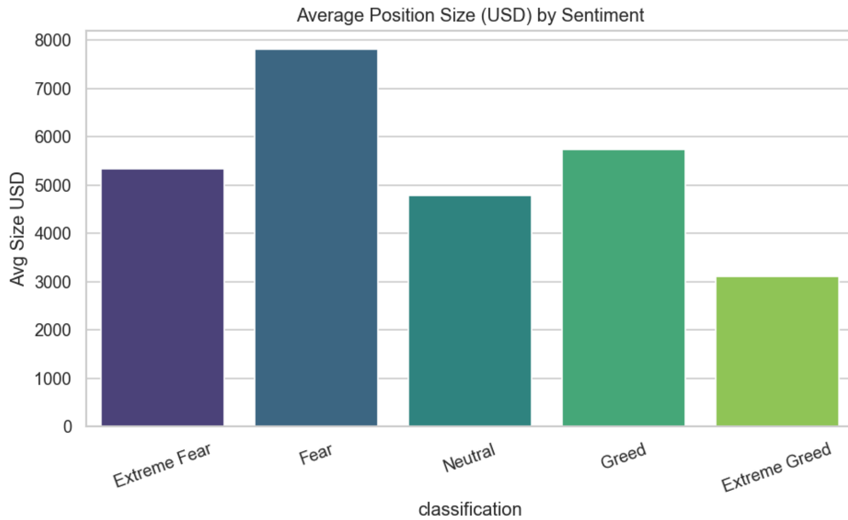

Figure 4. Average position size (USD) by sentiment regime.

3.7 Trader-level heterogeneity

Sentiment sensitivity is not uniform across the 32 accounts. A handful of accounts show large swings in average PnL between Fear and Greed regimes (e.g., account 0x420ab4… swings ~$3,194 avg PnL/trade between the two), while most accounts show smaller, single/double-digit swings — meaning the aggregate contrarian pattern is driven disproportionately by a subset of traders, not uniformly by all 32. (See `account_fear_vs_greed_pnl.csv` for the full ranked list.)

4. Statistical Significance

Descriptive differences are only useful if they're real, so every headline contrast was tested formally (non-parametric tests, since Closed PnL is heavy-tailed):

- **Trade-level Kruskal-Wallis** across all 5 regimes on Closed PnL: H = 730.33, **$p = 9.4\times10^{-157}$** — overwhelmingly significant, but trade-level tests over-state confidence because trades on the same day by the same account aren't independent observations (pseudo-replication).
- **Day-level Kruskal-Wallis** (the statistically correct unit of observation — one data point per calendar day) on daily total PnL across regimes: H = 20.06, **$p = 4.9\times10^{-4}$** — still significant, confirming the regime effect survives even at the conservative day-level granularity.
- **Chi-square test** of classification × win/loss independence: $\chi^2 = 1976.45$, dof = 4, **$p \approx 0$** — win rate is not independent of sentiment regime.

Pairwise comparison	Trade-level p (MW)	Day-level p (MW)	Win-rate z-test p
Fear vs Greed	6.7e-43	0.0095	~0
Extreme Fear vs Extreme Greed	4.6e-45	0.68 (n.s.)	~0
Extreme Fear vs Neutral	0.040	0.61 (n.s.)	~0
Fear vs Neutral	1.0e-36	0.43 (n.s.)	~0
Greed vs Extreme Greed	6.2e-116	0.000003	~0

Important nuance: at the day level (the correct unit), only *Fear vs Greed* and *Greed vs Extreme Greed* survive as statistically distinguishable in total daily PnL — the Extreme Fear vs Extreme Greed/Neutral contrasts, while dramatic-looking in the trade-level averages, are **not statistically distinguishable at the day level**, largely because Extreme Fear only spans 14 unique trading days in this dataset (see §7 Robustness). Win-rate differences, by contrast, are robustly significant everywhere.

5. Risk-Adjusted Performance

Higher average PnL can simply reflect bigger bets or bigger variance, not real edge — so each regime was re-evaluated on a risk-adjusted basis:

Sentiment	Trade Sharpe	Day Sharpe (ann.)	Day Sortino (ann.)	PnL/volume (bps)
Extreme Fear	0.044	8.28	25.73	64.6
Fear	0.084	6.06	14.06	69.5
Neutral	0.096	8.06	94.48	71.7
Greed	0.054	2.83	2.17	74.5
Extreme Greed	0.123	5.19	8.64	218.2

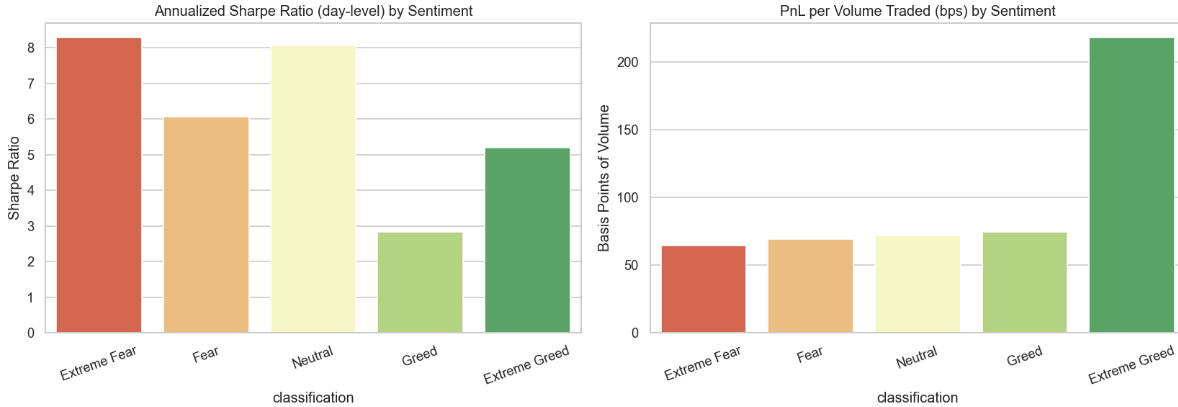

Figure 5. Annualized day-level Sharpe ratio and PnL-per-volume (bps) by sentiment regime.

- **Greed is the weakest regime risk-adjusted**, not just in absolute terms — it has both the lowest annualized Sharpe (2.83) and lowest Sortino (2.17) of all five buckets, meaning whatever edge exists there is thin relative to its volatility.
- **Extreme Greed is dramatically more capital-efficient**: 218 bps of PnL per dollar of volume traded — 3x any other regime — meaning traders are extracting far more profit per unit of risk/capital deployed during euphoric markets, not just trading bigger.
- **Neutral's Sortino (94.5) is the best of any regime**, driven by very few, very small down-days — it's a “low drama” regime: not the highest absolute return, but the smoothest.
- This reshapes the earlier win-rate-driven read: Extreme Greed remains the standout regime on every basis (return, win rate, *and* risk-adjusted efficiency), while Greed is confirmed as the weakest regime once risk is accounted for, not just an artifact of lower win rate.

6. Lagged / Predictive Analysis

Does yesterday's sentiment predict today's performance?

All results above are **contemporaneous** (same-day sentiment and outcomes) — not actionable for trading, since you can't act on a signal and observe its outcome on the same day. To test predictive value, each day's trading outcomes were conditioned on the **prior day's** sentiment classification:

Prior day's sentiment	Days (n)	Avg next-day PnL	Next-day win rate	Next-day volume
Extreme Fear	14	$72,701	84.9%	$9.15M
Fear	92	$34,751	81.6%	$5.05M
Neutral	64	$14,208	85.8%	$3.00M
Greed	198	$12,872	81.0%	$1.39M
Extreme Greed	111	$23,257	88.4%	$1.17M

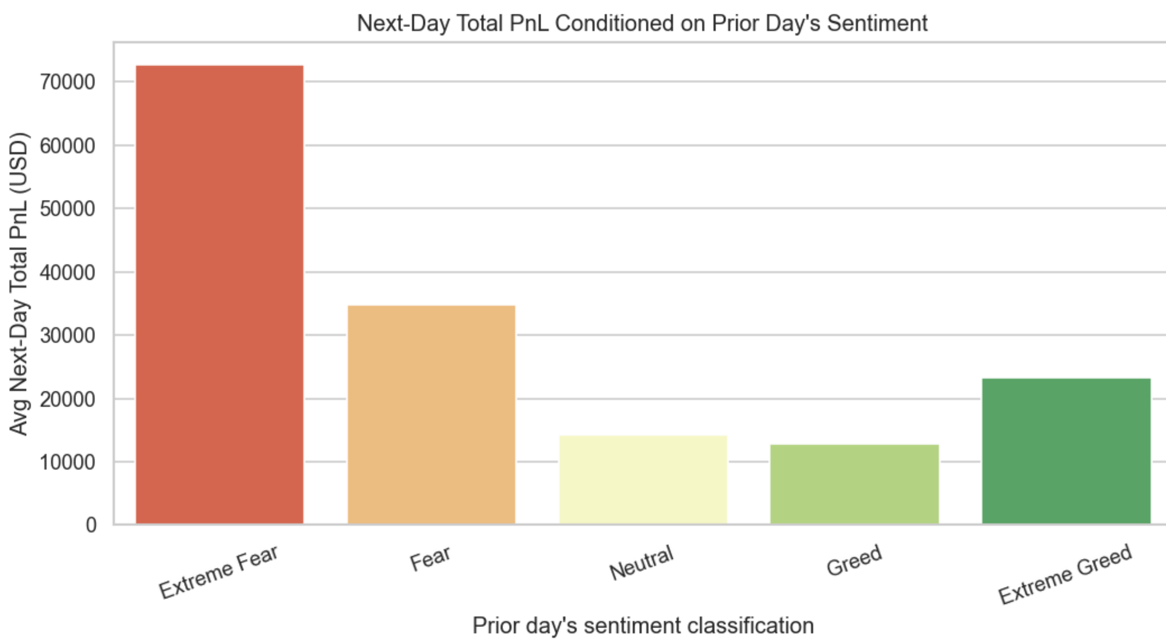

Figure 6. Average next-day total PnL conditioned on the prior day's sentiment classification.

This is the most actionable finding in the analysis. The day *after* an Extreme Fear reading produces the highest average next-day PnL of any condition in the entire study — over double the next-best (Fear→Fear) and roughly 5–6x the Greed/Neutral-conditioned days. A Kruskal-Wallis test on next-day PnL across prior-day regimes confirms this is statistically significant ($H = 22.36$, $\mathbf{p = 1.7\times10^{-4}}$).

Comparing same-day vs. next-day (lagged) correlation with the raw sentiment score:

Metric	Same-day corr.	Next-day (lagged) corr.
Total PnL	-0.083	-0.107
Total volume	-0.264	-0.277
Win rate	+0.055	+0.061

Metric	Same-day corr.	Next-day (lagged) corr.
Trade count	-0.245	-0.237

The lagged correlations are slightly *stronger* than the same-day ones for PnL and win rate — the predictive relationship is at least as strong as the contemporaneous one, meaning sentiment isn't just describing what already happened, it carries forward-looking signal for the next session.

Implication: the tradeable signal isn't “trade during Extreme Fear” (§3.3 showed that's actually the lowest-quality regime to trade *in*), it's “**position for the day after an Extreme Fear reading**” — likely capturing a fear-driven washout followed by a mean-reversion bounce.

7. Robustness Checks

Sentiment	Unique days	Raw avg PnL	Winsorized avg PnL	Top1 acct share	Top3 acct share
Extreme Fear	14	$71.03	$42.62	35.4%	84.9%
Fear	91	$112.63	$83.66	33.2%	62.6%
Neutral	67	$71.20	$57.19	31.0%	70.3%
Greed	193	$85.40	$65.78	24.8%	62.7%
Extreme Greed	114	$130.21	$92.59	40.7%	66.1%

- **Extreme Fear is the least robust regime**: only 14 unique days in the dataset, and the top 3 accounts account for **85% of its total PnL** — its results (both the weak win rate in §3.2 and the strong next-day effect in §6) should be treated as suggestive, not conclusive, until validated on more data.
- **Winsorizing (clipping the top/bottom 1% of trades) shrinks every regime's average PnL by 25–35%**, confirming outlier trades inflate the raw averages everywhere — but the *ranking* across regimes is preserved (Extreme Greed still highest, Extreme Fear/Neutral still lowest), so the qualitative conclusions are not an artifact of a few extreme trades.
- **PnL concentration in the top 1–3 accounts (25–41% / 63–85%) is high in every regime**, not just Extreme Fear — a reminder that this cohort's aggregate behavior is meaningfully whale-driven, and any strategy inspired by these findings should be validated against a broader trader sample before being generalized.

8. Strategic Implications for Trading

- **Fade the crowd, size for it.** The data supports a contrarian sentiment strategy: accumulate long exposure into Fear/Extreme Fear and rotate toward shorts/reduced exposure into Greed — this is what the most profitable segment of the historical flow is already doing.
- **The real edge is in the day *after* Extreme Fear, not during it.** Extreme Fear itself is a low-win-rate, low-Sharpe, statistically fragile (14-day) regime to trade in — but positioning for the session immediately following an Extreme Fear reading captures the strongest, statistically significant PnL effect in the whole dataset.
- **Extreme Fear is a volume trap.** It attracts the most trades but the lowest win rate — a signal to tighten entry criteria (wider stops, smaller clips, or wait for confirmation) rather than over-trade panic in real time.
- **Extreme Greed is the highest-quality regime on every measure** — return, win rate, Sharpe/Sortino, and capital efficiency (218 bps/volume, 3x any other regime). Allocate more conviction/capital to setups here; treat Greed (non-extreme) as the weakest risk-adjusted regime and reduce exposure there.
- **Use sentiment as a regime filter, not a standalone signal**, and prefer the lagged (previous-day) classification over the same-day one when sizing next-session risk — it carries equal-or-greater predictive power and is causally valid (you can act on yesterday's known reading).
- **Validate before productizing.** High top-3-account PnL concentration (63–85%) across every regime means these patterns are influenced by a handful of large traders. Before committing capital to a systematized version of this strategy, confirm the effect holds on a broader trader sample and out-of-sample time period.
- **Identify and study the sentiment-driven outperformers.** A small number of accounts contribute most of the fear/greed PnL spread; understanding their specific entries (coin choice, timing, leverage) could isolate a repeatable strategy worth productizing separately from the aggregate.

9. Caveats

- Closed PnL and win-rate figures are computed only on the 104k fills with non-zero Closed PnL; the other ~107k rows are position-opening/adjustment fills without a realized PnL and are excluded from those specific stats (but included in volume/direction stats).
- The dataset covers 32 accounts only; findings describe this specific trader cohort on Hyperliquid, not the broader market.
- No explicit leverage field existed in this data export; leverage effects are proxied via position size (Size USD) only.
- Sentiment is a single daily label for the whole market; it is a blunt instrument compared to intraday sentiment shifts, which this dataset can't capture.
- Extreme Fear's day-level results rest on only 14 unique calendar days — treat conclusions specific to that regime as directional, not definitive.
- Day-level Sharpe/Sortino figures annualize non-contiguous days within a regime (e.g., all “Fear” days treated as one time series) — a simplification for cross-regime comparability, not a claim that a real backtest would realize this exact ratio.

10. Files Delivered

- `analysis.py` — full reproducible analysis script
- `merged_trades_sentiment.csv` — trade-level data joined with daily sentiment
- `stats_by_sentiment.csv`, `daily_aggregates.csv`, `long_short_bias.csv`, `top_coins_by_sentiment.csv`, `account_performance_by_sentiment.csv`, `account_fear_vs_greed_pnl.csv`, `correlation_matrix.csv` — descriptive supporting tables
- `significance_tests.csv`, `significance_summary.txt` — statistical significance test results
- `risk_adjusted_metrics.csv` — Sharpe/Sortino/PnL-per-volume by regime
- `lagged_sentiment_analysis.csv`, `same_day_vs_lagged_correlation.csv` — predictive/lagged analysis
- `robustness_checks.csv` — sample size, concentration, and winsorization sensitivity checks
- `plot1–plot8 .png` — charts referenced above